

Material Modeling

Tutorial on using an Abaqus/CAE Plug-in to parse test data

Abaqus 6.11



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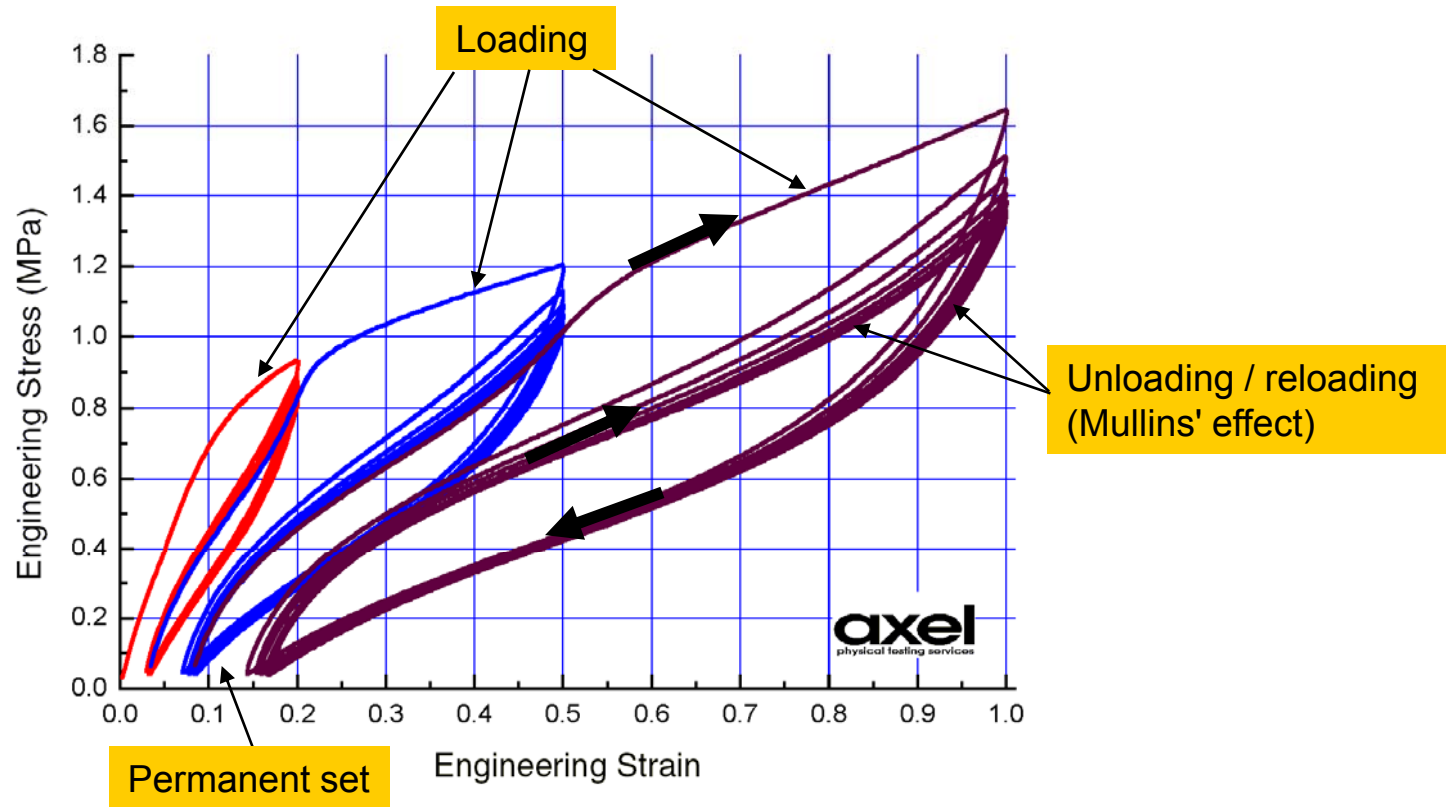


Outline

- **Motivation**
- **Approach**
- **Usage / Validation**
- **Summary**



Motivation: Test Data

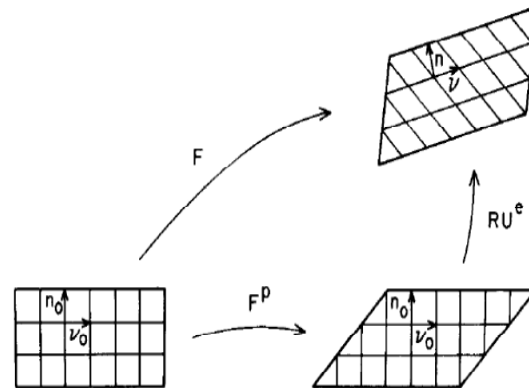


Goal: Ability to model the nonlinear loading, permanent set and, optionally, unloading / reloading (Mullins' effects)

Approach

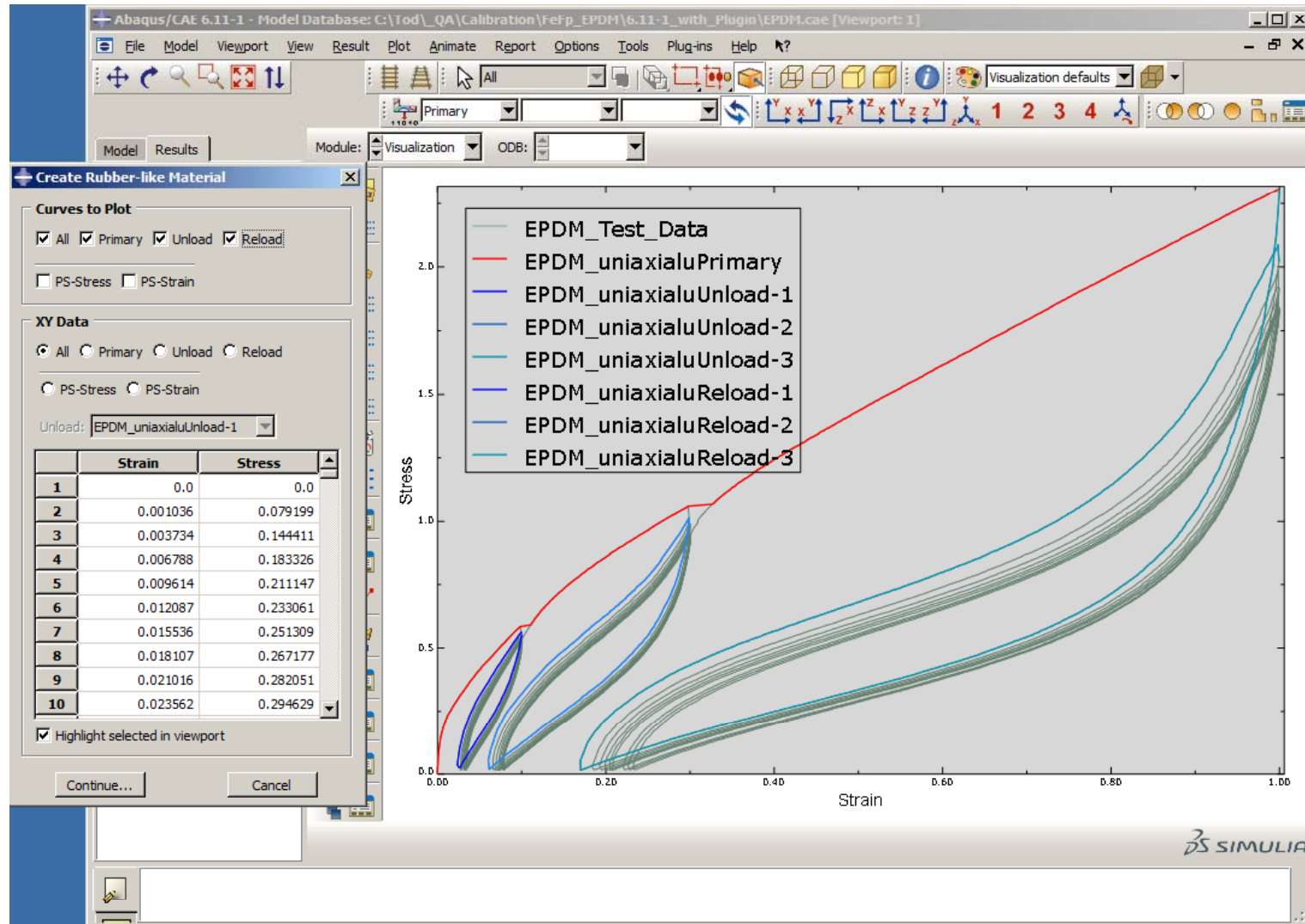
$$\mathbf{F} = \mathbf{F}^e \cdot \mathbf{F}^p$$

Lee (1969)



- ❑ Multiplicative split of the deformation gradient (motivated by crystal plasticity)
- ❑ Plasticity is modeled with isotropic hardening Mises plasticity
- ❑ Hyperelasticity can be modeled with any isotropic hyperelastic models available in Abaqus
- ❑ Can be combined with Mullins' effect to capture damaged response during unloading after initial loading

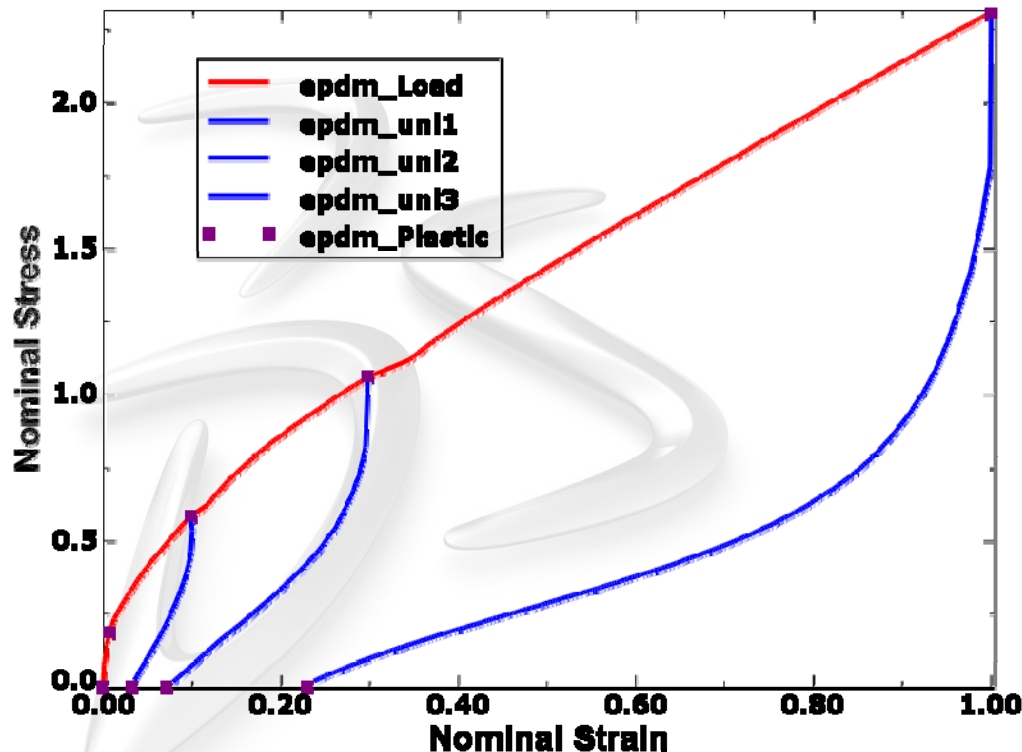
Usage, CAE GUI Plug-in – for Test Data Parsing



Usage, CAE GUI Plug-in – for Test Data Parsing

□ Characterize from experimental data using CAE plug-in

- The plug-in automatically extracts loading, unloading and permanent set data from uniaxial and biaxial test data
- Edit the data (remove any kinks, sudden jumps)
- Create a CAE material



Experimental data

CAE GUI
Test Data
Parsing

Loading, unloading
and permanent set

CAE Material (*Hyper
elastic, *Plastic, *Mullins)

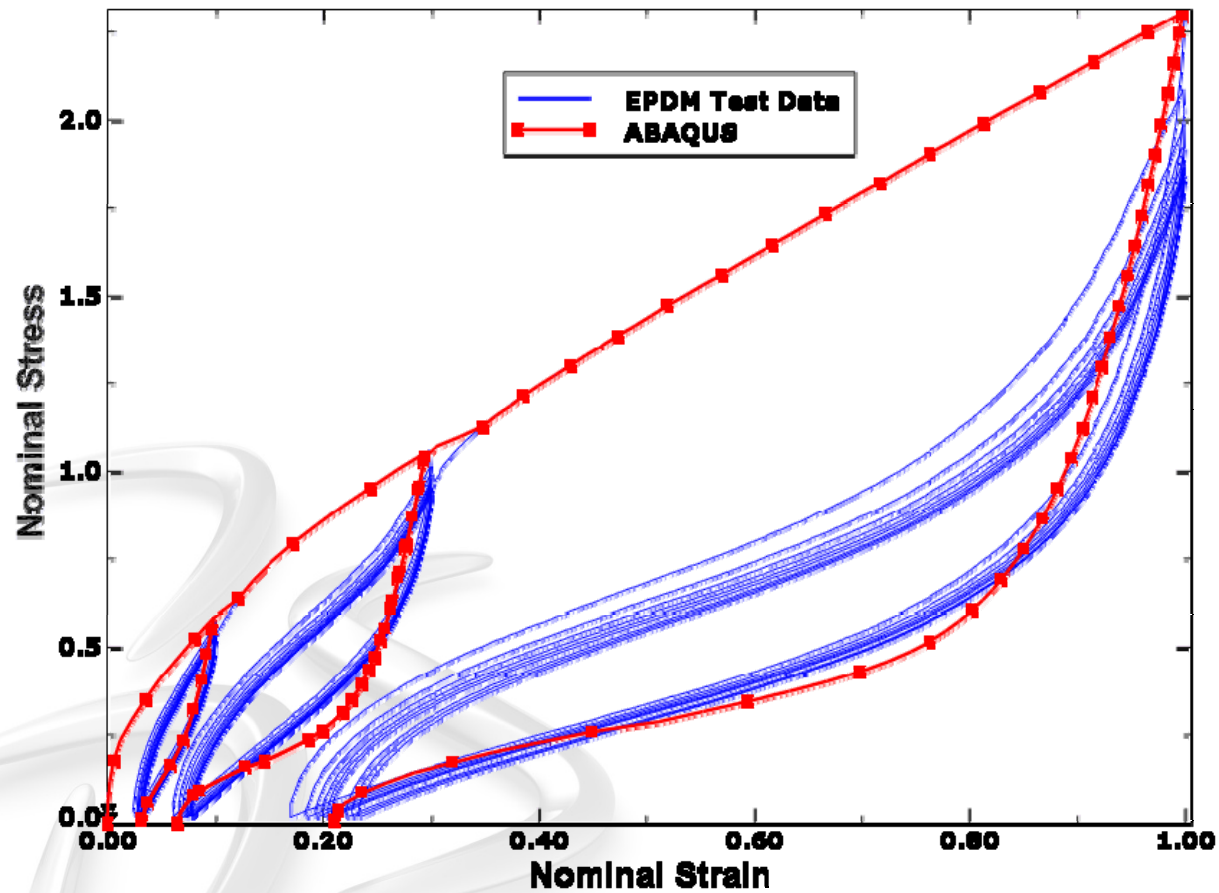
Usage, CAE GUI Plug-in – for Test Data Parsing

- Process two modes of test data, namely, uniaxial and biaxial.
- For each mode, the GUI will help users identify loading, permanent set and optionally, unloading / reloading data in their test data.
- Create data for the keywords: *Hyperelastic, Test Data, *Plastic and optionally, *Mullins Test Data
- Archive this information into a *Material
- The plug-in can be downloaded from SOSS Answer #3522

http://abaqus.custhelp.com/cgi-bin/abaqus.cfg/php/enduser/std_adp.php?p_faqid=3522&p_created=1193254530



Validation



□ Abaqus response from unit cube analysis, file st_epdm.inp

Summary

- **Multiplicative split of deformation gradient**
- **Following keywords can be combined**
 - ***HYPER ELASTIC**
 - ***MULLINS EFFECT** (optional)
 - ***PLASTIC, TYPE=ISOTROPIC**
- **Can be used to model filled elastomers and thermoplastics that show**
 - Permanent strain upon removal of load and/or
 - Damaged unloading behavior
- **Process test data through a CAE GUI plug-in**
- **Current Limitations**
 - Can not be combined with hyperelastic rate effects such as hysteresis or viscoelasticity
 - Will work with rate-independent isotropic hardening plasticity

Thank You

